



**UNIVERSITÉ
DE GENÈVE**

FACULTÉ DES SCIENCES
Section de physique

COSMOLOGY WITH FAST RADIO BURSTS

Student: S. Dubief

Supervision: A. Sampath

May 9, 2026

Contents

1	Introduction	2
2	Theory	3
2.1	What is a FRB?	3
2.2	DM: a fantastic tool	4
2.3	Macquart relation	5
2.3.1	Friedmann's equations and the Λ CDM model	5
2.3.2	The baryon density parameter	6
2.3.3	The cosmic dispersion measure	6
3	Methods	8
3.1	How can we detect FRBs?	8
3.2	Mock catalog	9
3.3	Cross-Matching	10
3.4	DM-z models	11
4	Results and discussion	11
4.1	Cross-Matching	11
4.2	Exploratory Plots	13
4.2.1	Histogram	14
4.2.2	Dispersion Measure for different models	14
5	Conclusion	16
6	Appendix	17
6.1	Errors on Residuals of DM_{IGM}	17
6.2	Variance	17
6.3	Evolution of Ω_b and f_{IGM}	17

1 Introduction

Fast Radio Bursts have rapidly become one of the most active topics in radio astronomy. The field is still very young: the first FRB was identified only in 2007 (Lormier & al., 2007, [1]), and was initially a surprising isolated event. Since then, the situation has changed a lot. With new wide-field radio telescopes, especially CHIME in Canada, the number of detected bursts has increased from a handful of sources to large statistical samples.



Figure 1: The CHIME radio telescope in British Columbia, Canada. Its wide field of view and continuous monitoring have made it one of the key instruments to study Fast Radio Bursts. (CHIME collaboration, 2026, [2])

The cosmological interest comes from the fact that their signals cross ionized matter before reaching Earth. Part of the observed dispersion measure is therefore produced by free electrons in the intergalactic medium, which makes FRBs useful tracers of baryons between galaxies.

This is particularly interesting because cosmology already gives a precise prediction for the total baryon density. Measurements of the cosmic microwave background (CMB), especially from the Planck mission, constrain the baryon content of the early Universe. However, at late times, a significant fraction of these baryons is expected to reside in the diffuse intergalactic medium, where it is hard to detect with traditional observations. FRBs therefore offer an independent and complementary test: by studying the relation between dispersion measure and redshift, one can ask whether the baryon content inferred today agrees with the value predicted from the early Universe.

In this project, we study whether FRBs can provide an independent late-time measurement of this baryon content. Using the Λ CDM framework, the Friedmann equation leads to the Macquart relation, which connects the intergalactic dispersion measure to redshift and to cosmological parameters such as H_0 , f_{IGM} , and Ω_b . We then compare this theoretical $DM - z$ relation with observational data to test whether the baryon density inferred from FRBs is consistent with Planck constraints, and to examine how the scatter may reflect large-scale structure along different lines of sight.

2 Theory

2.1 What is a FRB?

A Fast Radio Burst (FRB) is an extremely brief and powerful burst of radio wave energy coming from far outside our galaxy. These events last only a few milliseconds but release an enormous amount of energy in that short time, comparable to what the Sun emits over several days.

The exact origin of FRBs is still not fully understood, but many scientists think they are linked to highly energetic objects such as neutron stars (NS). Some FRBs repeat, indicating that their source can survive multiple outbursts, while others seem to occur only once, possibly due to more catastrophic events. Because of their extreme energy and distant origins, FRBs are considered one of the most mysterious objects in modern astrophysics.

FRBs are thought to originate either from isolated objects or from binary systems, with models favoring magnetars, which are NS characterized by extremely strong magnetic fields.

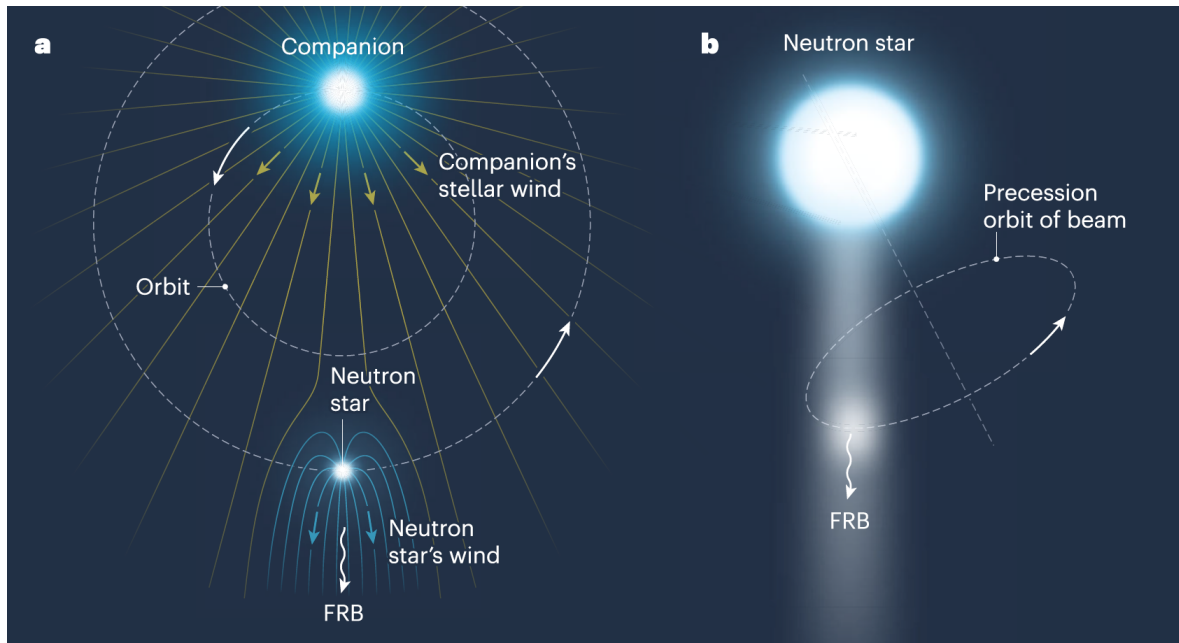


Figure 2: We distinguish two main types of possible FRB emitters: isolated objects and binary systems (Petroff & al., 2022, [3]).

For binary systems composed of at least one neutron star, there is a stronger emphasis on non-repeating or transient emission, depending on the exact mechanism. Pre-merger dynamics in NS–NS systems, as well as mergers or collisions involving compact objects (Totani, 2013, [4]), are generally associated with one-off FRBs due to their catastrophic nature, since the system is ultimately destroyed. In contrast, magnetic coupling or mass-transfer interactions in compact binaries could produce repeated or intermittent bursts, as long as the system remains intact and dynamically active (Petroff & al., 2022, [3]).

For single isolated objects, some models are consistent with star-forming galaxies, in particular a population of young magnetars (Kiyoshi & al., 2015, [5]; James & al., 2022, [6]). This is the case for shock models: radio emission is produced far from the central source when an ultra-relativistic outflow collides with the surrounding medium, generating a shock that emits through a synchrotron maser mechanism (Metzger & al., 2019, [7]). There are also magnetospheric models, which are coherent radio emission produced close to a neutron star when magnetic energy release generates disturbances that accelerate charges along magnetic field lines, leading to curvature radiation in the magnetosphere (Lu & al., 2020, [8]).

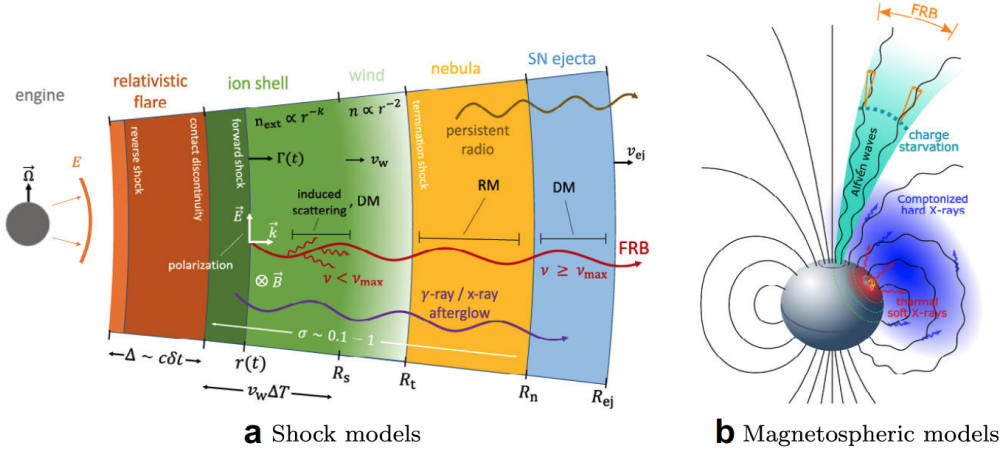


Figure 3: Schematic overview of two classes of FRB emission models. Left: shock models, in which the burst is produced at large radii from the compact central engine as a relativistic flare interacts with the surrounding medium. Right: magnetospheric models, in which the emission originates close to the neutron star, within its magnetosphere (Petroff & al., 2022, [3]).

2.2 DM: a fantastic tool

The dispersion measure is the integrated column density of free electrons along the line of sight to a radio source, causing lower frequency signals to arrive later than higher frequency ones due to their slower propagation through the plasma (Katz, 2025, [9]). In other words, this frequency-dependent delay arises from dispersion in the ionized medium. For a short-duration source at a distance D (units of pc), the DM is

$$DM = \int_0^D n_e dl \quad (1)$$

To measure the DM, we need to understand that the pulse travel time could be written in terms of the propagation time in vacuum and a delay due to the plasma. The latter arises

$$t_{pulse} = \frac{d}{c} + t_{delay} = \frac{d}{c} + \frac{e^2}{2\pi m_e c} \frac{DM}{\nu^2} \quad (2)$$

To find the time delay, we use the group velocity $v_g = \partial\omega/\partial k = dl/dt_d$ and the dispersion relation for an EM wave crossing a plasma $\omega^2 = \omega_p^2 + k^2 c^2$ (Maggiore, 2023, [10]), with the plasma frequency ω_p . Differentiating the time delay with respect to frequency ν (CASPER collaboration, 2021, [11]) gives

$$\frac{dt_d}{d\nu} = -\frac{e^2}{\pi m_e c} \frac{DM}{\nu^3} \quad (3)$$

Therefore, if we measure the rate of change of delay time with respect to the frequency, we can obtain the observed DM.

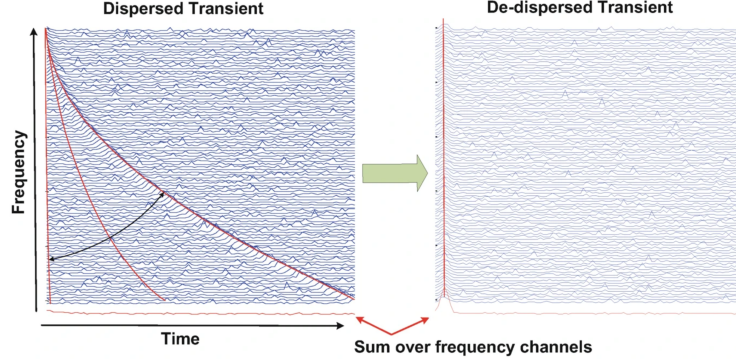


Figure 4: Left: Dynamic spectrum (time-frequency diagram) of a dispersed pulse from a pulsar or transient source, where lower frequencies arrive later, producing a characteristic sweep; the red curves illustrate trial dispersion measures that do not correctly align the signal. Right: After applying the correct de-dispersion, the signal is time-aligned across all frequencies (Schilizzi & al., 2024, [12]).

The measure of dispersion of a signal can be decomposed in several parts. The first contribution comes from the host galaxy and the local environment around the source. Although the intergalactic medium (IGM) is very low in density, it is the dominant contribution to the observed DM for sufficiently distant extragalactic FRBs. The last component is due to the Milky Way, including the free electrons along the line of sight from the Sun to the outer regions of the Milky Way (Xu & Han, 2015, [13]). Therefore, we can reasonably describe the observed DM (Macquart & al., 2020, [14])

$$DM_{obs} = DM_{host} + DM_{IGM} + DM_{MW}$$

2.3 Macquart relation

2.3.1 Friedmann's equations and the Λ CDM model

Our first task is to study the dynamics of the Universe in terms of the scale factor. We start with the Einstein field equations that relate the geometry of our Universe and its content

$$G_{\mu\nu} = 8\pi GT_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \quad (4)$$

where $T_{\mu\nu}$ is the energy-momentum tensor, $R_{\mu\nu}$ is the Ricci tensor and R is the Ricci scalar. The last term is obtained by the variational principle of the action.

Computing the Ricci tensor and the Ricci scalar, we obtain

$$G^0_0 = -3 \left(\left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right) = 8\pi(-\rho(t)) \quad ; \quad G^i_j = - \left(2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right) \delta^i_j \quad (5)$$

Now, with G^0_0 we can get the first Friedmann's equation

$$\left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3} \rho \quad (6)$$

Finally, to obtain the second Friedmann's equation, we take the trace of the spatial part of $G_{\mu\nu} = 8\pi T_{\mu\nu}$ and insert Eq. (6) into it, to obtain

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} (\rho + 3P) \quad (7)$$

We have assumed that our Universe is homogeneous and isotropic, so the energy-momentum tensor for a perfect fluid depends only on time. Note this tensor is written in terms of the pressure and the energy density. By the Bianchi identity we also assume covariant conservation of the energy-momentum tensor $\nabla_\mu T^\mu_\nu = 0$, which leads to the conservation equation.

The spatial curvature is consistent with zero, and its contribution to the expansion has always been subdominant compared to matter in the past and to the cosmological constant at late times, so we can set k to zero. In addition, radiation is also negligible today. Therefore

$$3H^2 = 8\pi G\rho = 8\pi G(\rho_m + \rho_\Lambda) \quad (8)$$

Dividing this equation by H_0^2 , using the definition of the energy density $\Omega_i = \rho_{i,0}/\rho_{c,0}$ with the critical density $\rho_{c,0} = 3H_0^2/8\pi G$, we obtain

$$H(z) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda} \quad (9)$$

where we also used the relations $1+z = a(t_0)/a(t_1)$ and $H = \dot{a}/a$.

2.3.2 The baryon density parameter

The second step is to write Ω_b in terms of the electron density in the IGM. On the one hand, we have

$$\Omega_b = \frac{\rho_{b,0}}{\rho_{c,0}} = \frac{\rho_b}{\rho_{c,0}} \frac{1}{(1+z)^3} \quad (10)$$

where the last equality comes from the conservation equation that we solve for baryons (i.e., $w = 0$). On the other hand, in a non-relativistic regime, the energy (or mass) density of baryons is given by

$$\rho_b = m_b n_b \simeq m_p n_b = m_p \frac{n_{e,\text{total}}}{\chi_e} \quad (11)$$

with n the number density of baryons or electrons and χ_e the free electron number per baryon. We can also assume that the IGM is only composed of hydrogen and helium. Now, because the number density of electrons is related to the total number density of electrons by a constant called f_{IGM} (Takahashi & al., 2021, [15]), we therefore have

$$\rho_b \simeq m_p \frac{n_e}{f_{\text{IGM}} \left(Y_H X_{e,H} + \frac{1}{2} Y_{He} X_{e,He} \right)} \quad (12)$$

where Y denotes the mass fraction of the considered elements and X_e the ionization fraction, both in the IGM. These values are given by (Lemos & al., 2023, [16]).

Finally, inserting this relation into (10)

$$\Omega_b = \frac{m_p}{\rho_{c,0}} \frac{n_e}{f_{\text{IGM}} \left(Y_H X_{e,H} + \frac{1}{2} Y_{He} X_{e,He} \right)} \frac{1}{(1+z)^3} \quad (13)$$

2.3.3 The cosmic dispersion measure

FRBs are often detected between $z = 0.2 - 1$, so we have to modify (1) to take into account the expansion. Indeed, the frequency is redshifted and time intervals are dilated at cosmological distances.

$$DM = \int_0^{z_{FRB}} \frac{n_e(z)}{1+z} \frac{c}{(1+z)H(z)} dz \quad (14)$$

Therefore, inserting (9) and using (13), the dispersion measure in the IGM is given by the Macquart relation (Macquart & al., 2020, [14])

$$\langle DM_{IGM} \rangle = \Omega_b \frac{3H_0 c}{8\pi G m_p} \int_0^{z_{FRB}} \frac{\left(Y_H X_{e,H} + \frac{1}{2} Y_{He} X_{e,He}\right) f_{IGM}(1+z)}{\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}} dz \quad (15)$$

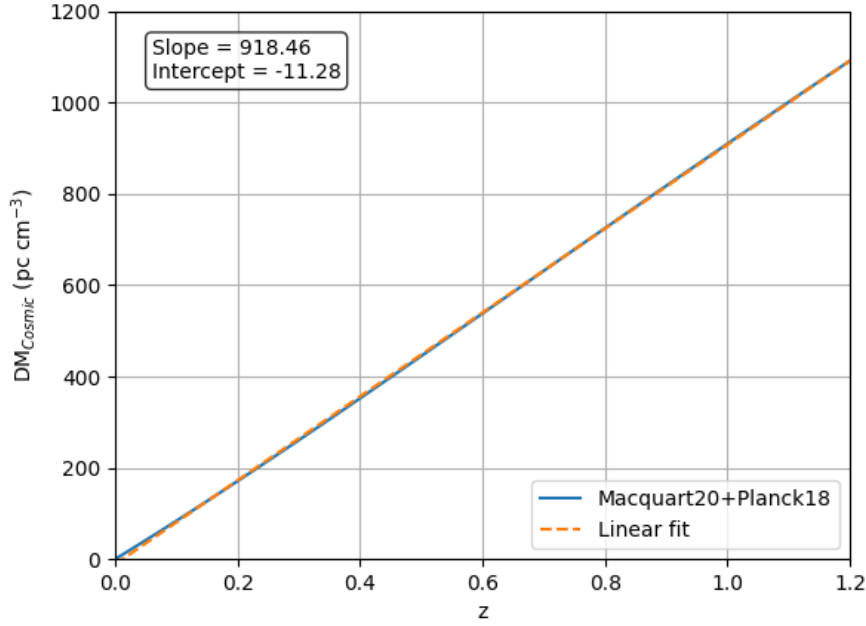


Figure 5: The dispersion measure as a function of redshift according to Eq. (15), shown in blue, calculated using the parameter values from (Planck collaboration, 2018, [17]).

The results indicate a linear relationship between the dispersion measure and redshift, reflecting the cumulative delay experienced by the signal as it propagates through ionized plasma.

The slope of this relation is proportional to the mean electron density in the low-redshift Universe for $z \lesssim 1$: $slope = dDM/dz \propto n_e \propto \Omega_b \propto \rho_b$. The intercept is consistent with zero within uncertainties, as expected in the absence of propagation. This supports the interpretation of DM as a measure of electron column density, confirming that FRBs can be used as effective and independent cosmological probes to constrain the baryonic content of the Universe.

Now, we vary some parameters in the Macquart relation and we can also plot the observed DM including the host contribution.

Varying cosmological parameters shows that changes in Ω_b or H_0 mainly rescale the overall amplitude of $DM(z)$, explaining why their effects can appear degenerate. In contrast, Ω_m modifies the expansion history through $H(z)$, altering the integrand and thus the shape of the relation. Since $DM \propto n_e$, it is directly sensitive to the free electron density, making Ω_b particularly influential. As a result, FRBs provide a probe of the ionized baryon content of the intergalactic medium, while their sensitivity to Ω_m arises indirectly through cosmological expansion.

For the observed dispersion measure, the host galaxy contribution must be included, typically of order $50\text{--}100 \text{ pc cm}^{-3}$ (Macquart & al., 2020, [14]). This is commonly modeled as

$$DM_{\text{obs}} = DM_{\text{IGM}} + \frac{DM_{\text{host}}}{1+z},$$

where the $(1+z)^{-1}$ factor accounts for cosmological time dilation and redshifting of the plasma effect. Contributions from the Milky Way may also be considered, but are often treated separately or subtracted using Galactic electron density models.

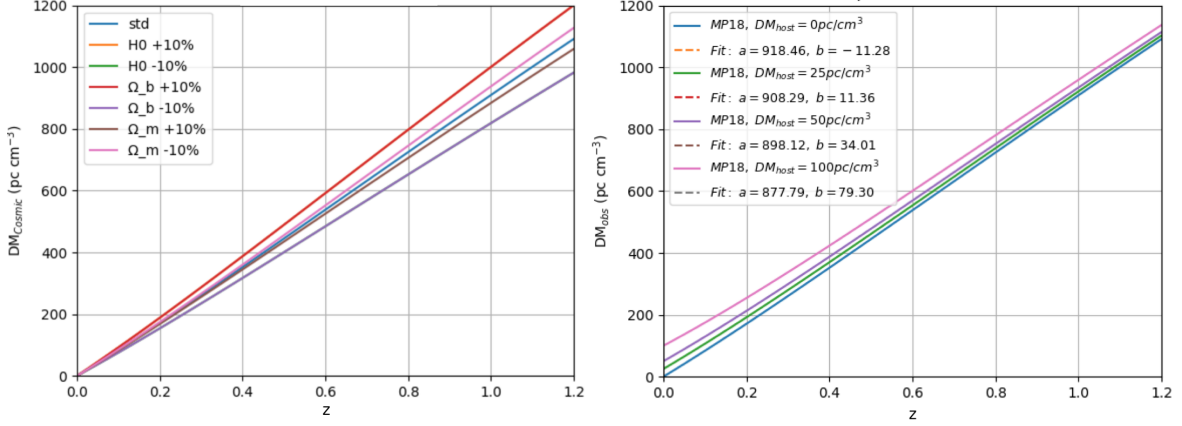


Figure 6: Left: variation of the cosmic dispersion measure when H_0 , Ω_b , and Ω_m are changed by $\pm 10\%$ around the standard cosmology. Right: observed dispersion measure including different host contributions. In the following analysis, we adopt $DM_{\text{host}} = 50 \text{ pc cm}^{-3}$.

3 Methods

3.1 How can we detect FRBs?

One of the most significant advantages of Fast Radio Bursts is that they can be detected from the ground. Radio waves are not significantly attenuated by Earth's atmosphere, making it far more practical to build radio telescopes than to deploy costly and difficult-to-repair satellites.

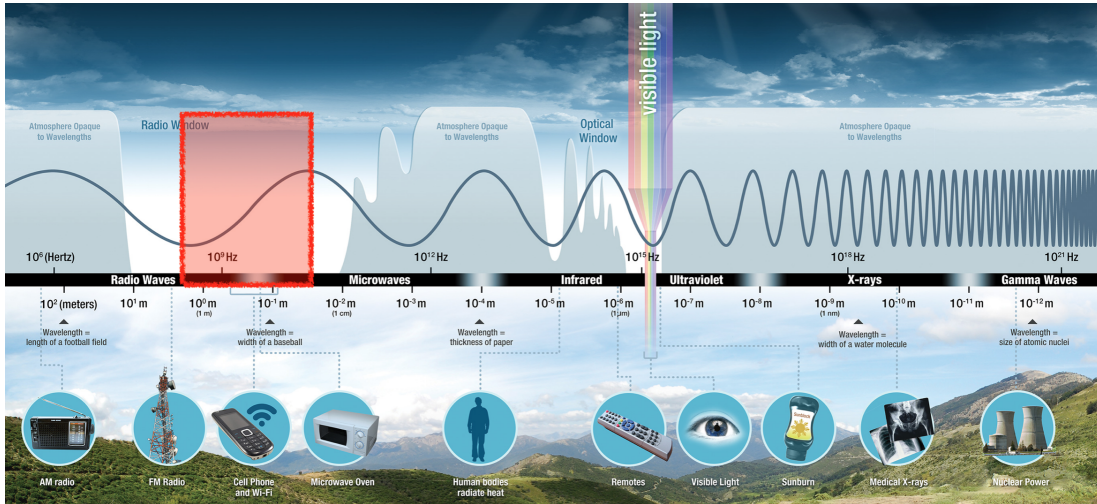


Figure 7: The red rectangle shows the possibility to detect FRBs from the Earth (NASA, 2010, [18]). They have been detected across a frequency range spanning 110 MHz to 8 GHz (Katz, 2025, [9]).

The radio telescopes in the Northern Hemisphere, such as CHIME and FAST, and those in the Southern Hemisphere, such as ASKAP and MeerKAT, together provide complementary sky coverage for the detection of Fast Radio Bursts. In this work, we will primarily use CHIME for observations in the Northern Hemisphere and ASKAP for those in the Southern Hemisphere in order to benefit from their complementary sky coverage.

CHIME is a Canadian wide-field radio telescope that is particularly efficient at detecting large numbers of FRBs due to its continuous monitoring of a broad portion of the sky. ASKAP is an array of radio antennas in Australia that offers a wide field of view and strong source localization capabilities, making it a key instrument for our study. Therefore, it is more interesting to rely on CHIME when investigating statistical properties such as population homogeneity or event rates. In contrast, when the goal is to accurately determine sky positions and identify potential host galaxies, we place greater confidence in ASKAP, thanks to its superior angular resolution and localization capabilities.

Finally, we will complement these observations with the FRBCAT, which is a compilation of FRBs detected by different ground-based radio telescopes worldwide such as Parkes and FAST.

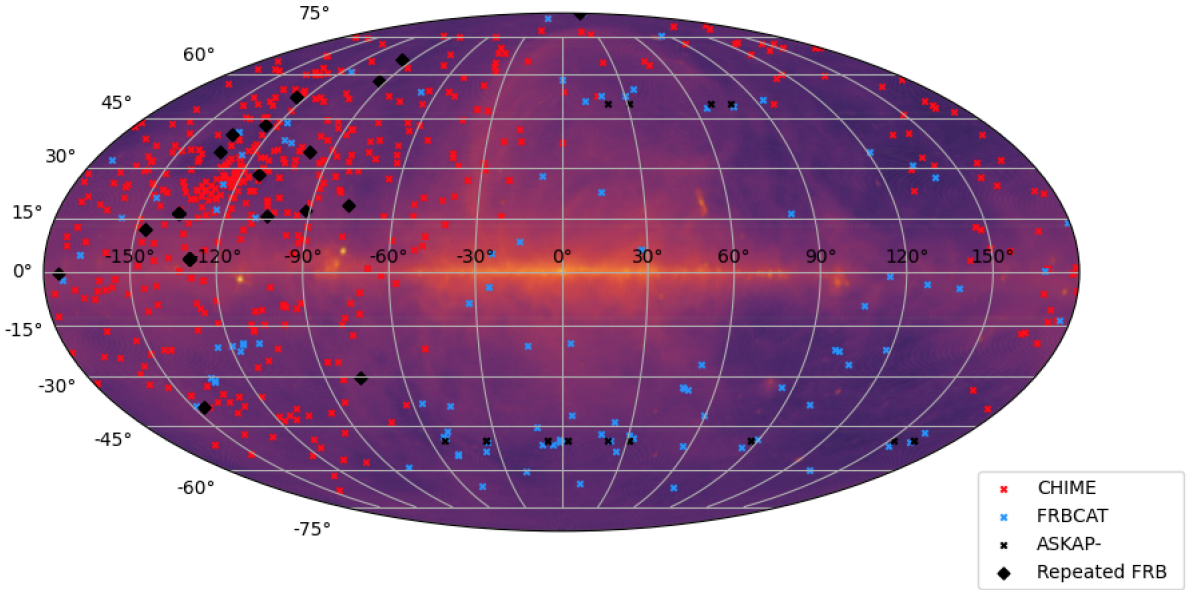


Figure 8: Following the section describing the different contributors to this graph, we used the CHIME catalog and FRBCAT, then added additional data from ASKAP already included in FRBCAT. We also identified repeated FRBs by checking for duplicates.

3.2 Mock catalog

Before using real data, we first construct a mock catalog based on the Macquart relation. This allows us to work with reliable simulated data in order to test our analysis methods.

Note that we directly plotted the observed dispersion measure. Indeed, adding the host contribution, $DM_{\text{host}}/(1+z)$, shifts the total dispersion measure upward relative to the pure intergalactic contribution, while its effect decreases with redshift because of the $(1+z)^{-1}$ factor. Thus, at low redshift the host contribution can be significant, whereas at higher redshift the observed relation is increasingly dominated by the IGM component.

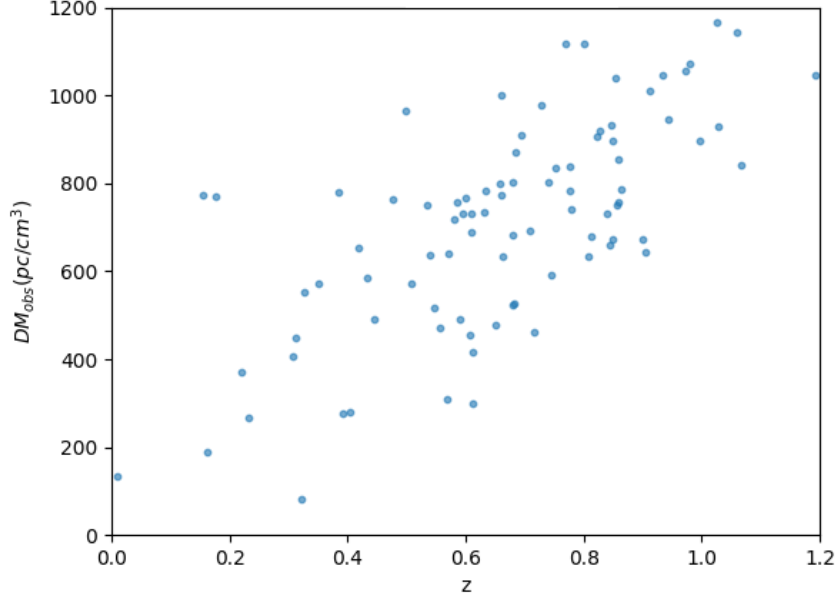


Figure 9: Example of a simulated FRB dataset.

3.3 Cross-Matching

An important step was to cross-match the FRB data from CHIME, FRBCAT, and ASKAP with SDSS and DESI in order to search for possible host-galaxy candidates associated with the FRBs.

We chose SDSS and DESI because they are large spectroscopic and photometric surveys containing many galaxy positions and redshifts. They are also complementary: SDSS provides well-established and widely used catalog data, while DESI extends the available coverage with more recent and deeper observations.

Our main difficulty was the large uncertainty in right ascension and declination in the CHIME catalog, which could cause some possible host candidates to be missed. Conversely, the cross-match can also include false positives.

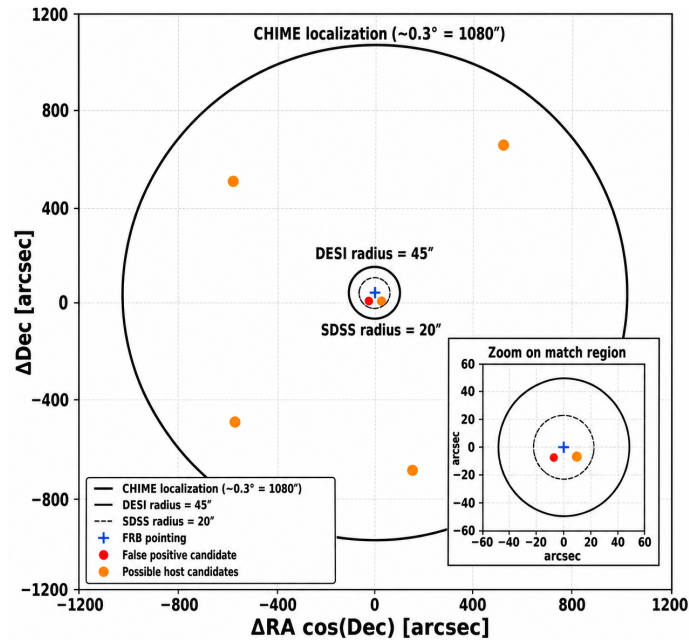


Figure 10: Illustration of the FRB cross-match region. The large circle shows the CHIME localization uncertainty, which can reach $\sim 0.3^\circ$, while the smaller circles show the adopted SDSS ($20''$) and DESI ($45''$) search radii. The horizontal axis is written as $\Delta\text{RA} \cos(\text{Dec})$ because angular separations in right ascension are reduced by this factor on the sky, preventing a geometric distortion of the circular search regions. Orange points illustrate possible catalog targets, including objects inside the CHIME localization region but outside the SDSS/DESI cross-match radii, which could correspond to missed host candidates. The red point indicates a known galaxy within the search radius that is not the true FRB host, representing a false positive.

We can also assume a relatively low error for the FRBCAT and the additional ASKAP catalog, because they mostly include FRBs with more precise sky localizations than the CHIME catalog. In particular, ASKAP detections often benefit from interferometric localization, which can constrain the FRB position much more accurately and therefore reduce the risk of missing the true host during the cross-match (Bannister & al., 2019, [19]).

3.4 DM- z models

The Macquart relation provides a good linear model $DM_{IGM} = az + b$. However, we will compare it with a quadratic model $\log DM_{IGM} = a(\log z)^2 + b(\log z) + c$ to check whether our theoretical development is consistent (Piratova-Moreno & Garcia, 2024, [20]). This quadratic model is motivated by numerical simulations of the intergalactic medium, which suggest that the $DM_{IGM} - z$ relation may deviate from a simple linear trend. FRBs at similar redshifts can have very different DMs, meaning that a purely radial model $DM(z)$ may miss part of the relevant physics.

4 Results and discussion

4.1 Cross-Matching

After letting our algorithm find the matches between the FRBs and the possible host galaxies, the results are compiled in a table on the following page.

The first interesting point is that, with the angles defined in the previous section, we have enough matches to test the consistency with our theoretical relation (15). Moreover, only one candidate was identified for each FRB, which avoids the need for manual selection for most sources, although this would still be possible in some specific cases. Nevertheless, we had to remove one candidate because it had a redshift of -0.000411 and a DM of 300 pc cm^{-3} , which is not consistent.

Since the candidates are relatively distant, we detected only galaxies, with the exception of one quasi-stellar object (QSO). A normal galaxy is dominated by starlight, whereas a QSO is a galaxy whose emission is dominated by an extremely luminous active galactic nucleus (AGN) powered by accretion onto a supermassive black hole. Another interesting point is the presence of two star-forming galaxies, which supports what we discussed in the theoretical section: some FRB models are consistent with environments associated with active star formation.

Thanks to the cross-match result, can conclude that the late-time baryon density appears broadly consistent with CMB measurements. The CMB predicts the total amount of baryonic matter expected in the early Universe, while FRBs probe where part of this matter is today through the ionized IGM. Since DM generally increases with redshift in our sample, it suggests that many of the baryons predicted by the CMB are still present at late times in the ionized IGM.

FRB name	α	$\Delta\alpha$	δ	$\Delta\delta$	DM_{obs}	ΔDM_{obs}	z	Δz	Class	Subclass	Catalog
FRB20180727A	197.7153	0.13	26.4246	0.26	642.1340	0.0150	0.5466	1.0135e-4	GALAXY	–	DESI
FRB20180923D	169.0726	0.02	48.7498	0.07	329.3996	0.0020	0.5795	1.4555e-4	GALAXY	–	SDSS
FRB20181213B	183.5083	0.22	53.7059	0.23	626.5930	0.0150	0.3518	6.9990e-5	GALAXY	–	DESI
FRB20181214D	178.5644	0.20	46.7096	0.19	1177.3190	0.0180	0.5601	1.3531e-4	GALAXY	–	SDSS
FRB20190103D	221.8342	0.24	59.9143	0.24	1913.5509	0.0086	0.4062	8.3172e-5	GALAXY	–	SDSS
FRB20190202B	114.3125	0.039	32.4410	0.18	464.9060	0.0022	0.2163	3.6785e-5	GALAXY	–	DESI
FRB20190208C	141.5594	0.037	83.5522	0.22	238.3921	0.0024	0.1578	8.5769e-5	GALAXY	–	DESI
FRB20190212B	139.9866	0.18	52.1140	0.17	600.1866	0.0028	0.2180	2.1234e-5	GALAXY	STARFORMING	SDSS
FRB20190222C	239.1788	0.04	40.0298	0.19	524.0071	0.0066	0.1969	2.9810e-5	GALAXY	–	SDSS
FRB20190409D	336.6397	0.23	26.9476	0.21	1300.1180	0.0110	0.9098	3.4915e-4	QSO	–	SDSS
FRB20190601A	190.1340	0.21	62.7145	0.21	2227.8910	0.0120	1.0015	2.5299e-5	GALAXY	–	DESI
FRB20190604E	235.5938	0.21	32.0489	0.21	1218.5965	0.0077	1.4564	2.2565e-4	GALAXY	–	DESI
FRB20190617C	134.3722	0.21	35.6961	0.21	640.1560	0.0230	0.4600	1.7168e-4	GALAXY	–	SDSS
FRB190608	334.0204	nan	-7.8989	nan	338.7000	0.5000	0.1178	8.2585e-6	GALAXY	STARFORMING	SDSS
FRB190303.J1353+48 [†]	208.2480	nan	48.2505	nan	222.4000	0.2000	0.1674	3.3999e-5	GALAXY	–	SDSS
FRB190116.J1249+27 [†]	192.2516	nan	27.1467	nan	444.0000	0.6000	0.1782	3.2991e-5	GALAXY	–	DESI
FRB180525	220.0098	nan	-2.2066	nan	388.1000	0.3000	0.7852	1.0173e-4	GALAXY	–	DESI
FRB151125	22.7536	nan	30.9750	nan	273.0000	4.0000	0.6449	5.5605e-5	GALAXY	–	DESI

Table 1: FRB galaxy catalog obtained from our cross-match. The daggers indicate repeaters, while NaN values denote quantities that were not available in the databases. Interestingly, two of the matched FRBs are associated with star-forming galaxies, in agreement with the discussion presented in the theoretical section.

4.2 Exploratory Plots

In the theoretical section, we had visualized a set of FRBs in a Galactic view. This allowed us to observe a certain homogeneity in these radio signals, despite detection biases.

If we examine the repeaters in this Mollweide projection, we notice a correlation with the candidate table from our cross-match. Indeed, two candidates in this table are repeaters: FRB190303.J1353+48, with 17 bursts (Michilli & al., 2023, [21]), and FRB190116.J1249+27, with 2 bursts (CHIME Collaboration, 2019, [22]).

The first source, also known as FRB20190303A, is associated with a luminous and massive spiral galaxy undergoing significant star formation at $z \simeq 0.064$ (Michilli et al., 2023, [21]), although the association remains uncertain. Indeed, our inferred value for this FRB is $z = 0.1674$, with a very small uncertainty, which is significantly higher than $z \simeq 0.064$. This raises the possibility that the third candidate reported in their field, with a photometric redshift of $z = 0.20 \pm 0.04$, could be more consistent with our result. Taking this uncertainty into account, our localization in Table 1 would therefore favor this third candidate as a plausible alternative counterpart for FRB190303.J1353+48.

As for the second repeater candidate, FRB190116.J1249+27, it appears not to be associated with any known object.



Figure 11: If we query the coordinates of FRB190303.J1353+48 from Table 1 in SIMBAD, they appear to coincide with a galaxy. However, it cannot be ruled out that the nearby object next to this galaxy is the actual emitter.

The low observed repeater fraction, 2.67%, may indicate that a large fraction of FRBs are produced by cataclysmic events. However, this fraction is also strongly affected by observational biases, since some apparently non-repeating FRBs may simply not have been observed long enough or with sufficient sensitivity to reveal repeated bursts. Note also that possible progenitors of repeating FRBs, such as magnetars, may be relatively uncommon or active only during a short phase, which could contribute to the small number of detected repeaters.

The fraction of localized candidates relative to the total number of FRBs is exactly the same. This suggests that the limiting factor is not necessarily the FRB sample itself, but rather the host-identification procedure. This may be due to the incompleteness and limited sky coverage of the galaxy catalogs used here, namely SDSS and DESI. This result may also be affected by our cross-matching selection bias, as illustrated in Figure 10. In particular, we adopted conservative selection criteria in order to ensure reliable associations, even if this reduces the number of selected candidates. Other possible explanations include incompleteness at high redshift or for faint host galaxies, as well as uncertainties in the FRB localization regions. Indeed, at redshifts $z \gtrsim 1$, FRB detection becomes more challenging because of their lower observed flux, larger dispersion measures, and possible temporal broadening due to scattering. These effects can reduce the detectability of distant FRBs and contribute to the incompleteness of the sample.

4.2.1 Histogram

The DM distribution is mainly concentrated at low to intermediate values, with most FRBs found between about 200 and 800 pc cm^{-3} , while a smaller number extends toward much higher DM. The lack of events below 100 pc cm^{-3} should not be over-interpreted, since CHIME analyses often use a $\text{DM} > 100 \text{ pc cm}^{-3}$ cut to select likely extragalactic sources. Repeaters also appear to be fewer and mostly located at lower DM in our histogram. However, this trend should be interpreted with caution (Amiri & al., 2021, [23]): since there are fewer detections in the high-DM tail, it is also expected to find fewer repeaters there. In our case, the sample size is not large enough to draw a firm conclusion from this apparent difference.

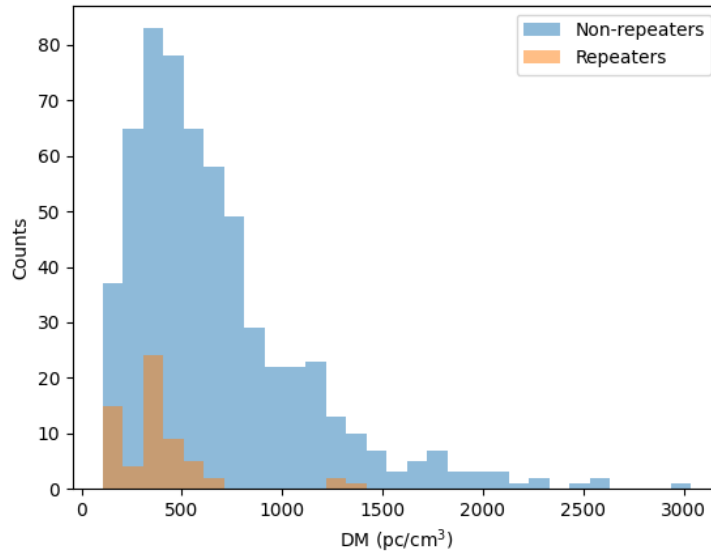


Figure 12: The DM data come from CHIME and FRBCAT, while the ASKAP data were not provided.

4.2.2 Dispersion Measure for different models

In the simplest picture, DM_{IGM} is expected to scale almost linearly with redshift at low z . However, departures from a perfect linear relation are expected because each line of sight crosses a different environment, especially overdense regions such as filaments. At low redshift, the inferred contribution also depends strongly on the direction through our Galaxy.

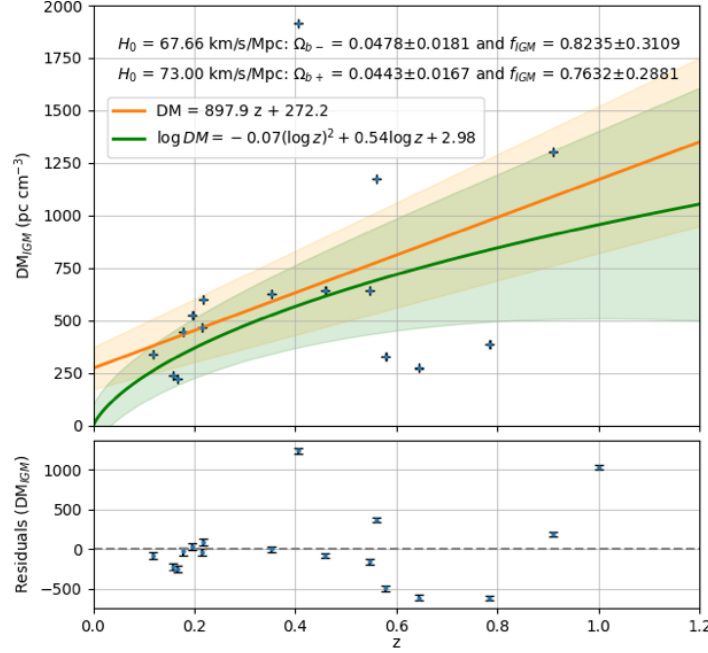


Figure 13: The figure shows the evolution of the intergalactic dispersion measure as a function of redshift for the FRB sample. The general trend is clearly increasing: more distant FRBs cross a larger column of ionized baryons in the IGM, producing a larger dispersion measure. The coefficients of the fits were obtained using a least-squares polynomial fit. The uncertainties in Table 1 mainly correspond to instrumental or catalog-level errors for the selected candidates. They are small on the scale of the plot, so the error bars are almost invisible. The lower panel shows the residuals with respect to the linear model, which is not really different from the quadratic log model.

Two different empirical models are compared in the figure. The first one is a linear relation shown in orange, and the second is a logarithmic quadratic model shown in green. Both models differ mostly at low ($z < 0.2$) and intermediate redshift ($z > 0.8$). The linear model is simpler and directly related to the usual theoretical expectation for the mean cosmic baryon density, while the logarithmic quadratic model allows more curvature and can partly absorb deviations caused by the scatter of the data.

The residuals are defined as $r_i = DM_{IGM,i} - DM_{model,i} = DM_{IGM,i} - (az_i + b)$. They are large for several FRBs, sometimes reaching several hundred pc cm^{-3} , which shows that the scatter is not dominated by the small measurement uncertainties. The residuals obtained with the quadratic logarithmic model are very similar, meaning that the present sample does not strongly favor one model over the other. To see how we have determined the error bars on these residuals, the calculation is given in the Appendix. The physical interpretation of the residuals is also important. Following the idea in the first paragraph, FRBs do not all propagate through the same type of cosmic environment. The scatter around the fitted curve is thus connected to the distribution of baryons in the large-scale structure (LSS). Since this structure grows from primordial density perturbations, the $DM - z$ relation is indirectly linked to the history of structure formation and, at a more fundamental level, to the initial perturbations generated during inflation.

We do not fit Ω_b and f_{IGM} simultaneously. Instead, one parameter is fixed while the other is inferred from the slope of the $DM - z$ relation. Using the parameter values of (Planck collaboration, 2018, [17]), the fit gives $\Omega_b = 0.0478 \pm 0.0181$ and $f_{IGM} = 0.8235 \pm 0.3109$, which is consistent with the standard expectation that a large fraction of baryons is located in the diffuse IGM. To visualize how Ω_b and f_{IGM} behave as functions of H_0 , see the Appendix.

The uncertainty on Ω_b or on f_{IGM} is mainly obtained from the uncertainty on the slope of the linear fit; this is why the errors are quite large. The function used for the fit (i.e., `np.polyfit()`) estimates the error on the slope from the dispersion of the points around the fitted line. Since the residuals are sometimes very large, the uncertainty on the slope also becomes large. Therefore, the error bars on Ω_b and f_{IGM} reflect the astrophysical scatter of the FRB sample but not the instrumental uncertainty on each individual point.

To compare the two models more quantitatively, we compute the median absolute error $\hat{x}$ and the root mean square error (RMSE). These metrics are evaluated using the observed dispersion measure DM_{obs} , by comparing it to the total predicted dispersion measure calculated with our models of DM_{IGM}

$$DM_{\text{obs,pred}} = DM_{IGM,model} + DM_{MW} + \frac{DM_{host}}{1+z} \quad (16)$$

The median absolute error measures the typical discrepancy between the predicted and observed DM and is robust against outliers. The RMSE gives more weight to large residuals and is therefore more sensitive to strongly discrepant data points.

	Median Error $\hat{x}$	RMSE
Formulas	$\text{median}(DM_{\text{obs},i} - DM_{\text{obs,pred},i})$	$\sqrt{\frac{1}{n} \sum_i (DM_{\text{obs},i} - DM_{\text{obs,pred},i})^2}$
Linear	261.7	482.4
Quadratic log	175.6	478.3

Table 2: Comparison of the performance metrics for the two models. Here, $n = 18$ denotes the number of data points.

The logarithmic quadratic model provides a better median absolute error than the linear model, suggesting a better typical agreement with the data. However, the RMSE values are very similar, indicating that large residuals or outliers still dominate the overall scatter.

5 Conclusion

In this work, we studied how Fast Radio Bursts can be used as probes of the baryonic content of the late Universe. The general increase of DM_{IGM} with redshift is consistent with the idea that more distant bursts cross a larger column of ionized matter.

From the linear fit, we separately obtained $\Omega_b = 0.0478 \pm 0.0181$ and $f_{IGM} = 0.8235 \pm 0.3109$, which are compatible with the standard cosmological picture and with Planck constraints (CMB). This suggests that FRBs can provide an independent late-time measurement of the baryon content, although the present sample is still too small and scattered to give a precise constraint.

We also compared the linear Macquart-inspired model with a logarithmic quadratic model. The latter gives a smaller median absolute error, but the RMSE remains nearly unchanged, showing that large residuals still dominate the scatter. The linear model remains the most natural interpretation for cosmology, while the quadratic model is useful only as a phenomenological comparison. Overall, our results support the potential of FRBs as cosmological probes, but also highlight the need for larger samples of well-localized FRBs with reliable host-galaxy redshifts.

6 Appendix

6.1 Errors on Residuals of DM_{IGM}

The relation for the cosmic DM is $DM_{IGM} = DM_{\text{obs}} - DM_{\text{MW}} - \frac{DM_{\text{host}}}{1+z}$, so the error is

$$\sigma_{DM_{IGM}}^2 = \left(\frac{\partial DM_{IGM}}{\partial DM_{\text{obs}}} \sigma_{DM_{\text{obs}}} \right)^2 + \left(\frac{\partial DM_{IGM}}{\partial DM_{\text{MW}}} \sigma_{DM_{\text{MW}}} \right)^2 + \left(\frac{\partial DM_{IGM}}{\partial DM_{\text{host}}} \sigma_{DM_{\text{host}}} \right)^2 + \left(\frac{\partial DM_{IGM}}{\partial z} \sigma_z \right)^2$$

$$\Rightarrow \sigma_{DM_{IGM}} = \sqrt{\sigma_{DM_{\text{obs}}}^2 + \sigma_{DM_{\text{MW}}}^2 + \left(\frac{\sigma_{DM_{\text{host}}}}{1+z} \right)^2 + \left(\frac{DM_{\text{host}} \sigma_z}{(1+z)^2} \right)^2}.$$

Finally, we neglect σ_z and $\sigma_{DM_{\text{MW}}}$, so we obtain $\sigma_{DM_{IGM}} = \sqrt{\sigma_{DM_{\text{obs}}}^2 + \left(\frac{\sigma_{DM_{\text{host}}}}{1+z} \right)^2}$.

6.2 Variance

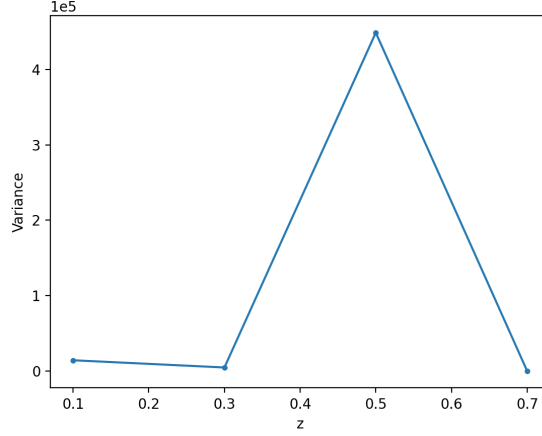


Figure 14: The residual variance was computed in redshift bins. Because the number of data points per bin is limited and unevenly distributed, the variance estimate is noisy and strongly affected by outliers. No robust monotonic trend with redshift can be inferred from this binned variance alone.

6.3 Evolution of Ω_b and f_{IGM}

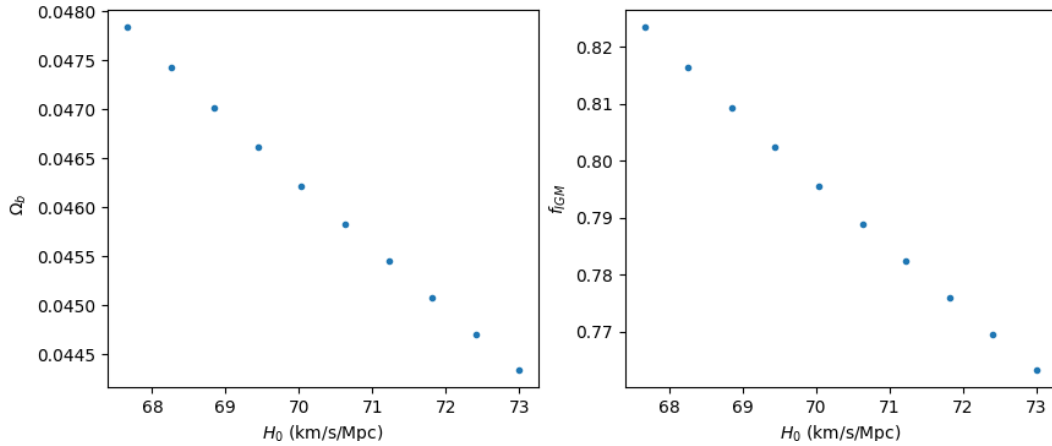


Figure 15: Evolution of baryonic parameters. As H_0 increases, the inferred values of Ω_b and f_{IGM} decrease because a faster cosmic expansion reduces the amount of baryonic matter required to reproduce the same observed signal.

References

- [1] D. R. Lormier & al., A Bright Millisecond Radio Burst of Extragalactic Origin, *Science* 318, 2007.
- [2] CHIME Collaboration, CHIME Media, 2026.
- [3] E. Petroff & al., Fast radio bursts at the dawn of the 2020s, *A&A* 30, 2022.
- [4] T. Totani, Cosmological Fast Radio Bursts from Binary Neutron Star Mergers, arXiv 1307.4985, 2013.
- [5] M. Kiyoshi & al., Dense magnetized plasma associated with a fast radio burst, *Nature* 528, 2015.
- [6] C. W. James & al., A measurement of Hubble's Constant using Fast Radio Bursts, *MNRAS* 516, 2022.
- [7] B. D. Metzger & al., Fast radio bursts as synchrotron maser emission from decelerating relativistic blast waves, *MNRAS* 485, 2019.
- [8] W. Lu & al., A unified picture of Galactic and cosmological fast radio bursts, *MNRAS* 498, 2020.
- [9] J. I. Katz, Fast Radio Bursts, arXiv 2510.19143, 2025.
- [10] M. Maggiore, A Modern Introduction to Classical Electrodynamics, Oxford, 2023.
- [11] CASPER collaboration, Dispersion measure, AstroBaki, 2021.
- [12] R. T. Schilizzi, Innovation Meets Reality: The SKA Design, Springer Nature, 2024.
- [13] J. Xu & J. L. Han, Extragalactic dispersion measures of fast radio bursts, arXiv 1504.00200, 2015.
- [14] J.-P. Macquart & al., A census of baryons in the Universe from localized fast radio bursts, *Nature* 581, 2020.
- [15] R. Takahashi & al., Statistical modelling of the cosmological dispersion measure, *MNRAS* 502, 2021.
- [16] T. Lemos & al., Forecasting constraints on the baryon mass fraction in the IGM from fast radio bursts and type Ia supernovae, Springer Nature, 2023.
- [17] Planck collaboration, Planck 2018 results. VI. Cosmological parameters, *A&A* 641, 2018.
- [18] NASA Introduction to the Electromagnetic Spectrum, NASA Science, 2010.
- [19] K. W. Bannister & al., A single fast radio burst localized to a massive galaxy at cosmological distance, arXiv 1906.11476, 2019.
- [20] E. F. Piratova-Moreno & L. A. Garcia, Modeling the dispersion measure—redshift relation for fast radio bursts, *Frontiers*, 2024.
- [21] D. Michilli & al., Subarcminute Localization of 13 Repeating Fast Radio Bursts Detected by CHIME/FRB, *ApJL* 950, 2023.

-
- [22] CHIME Collaboration, CHIME/FRB Detection of Eight New Repeating Fast Radio Burst Sources, arXiv 1908.03507, 2019.
- [23] M. Amiri & al., The First CHIME/FRB Fast Radio Burst Catalog, ApJS 257, 2021.